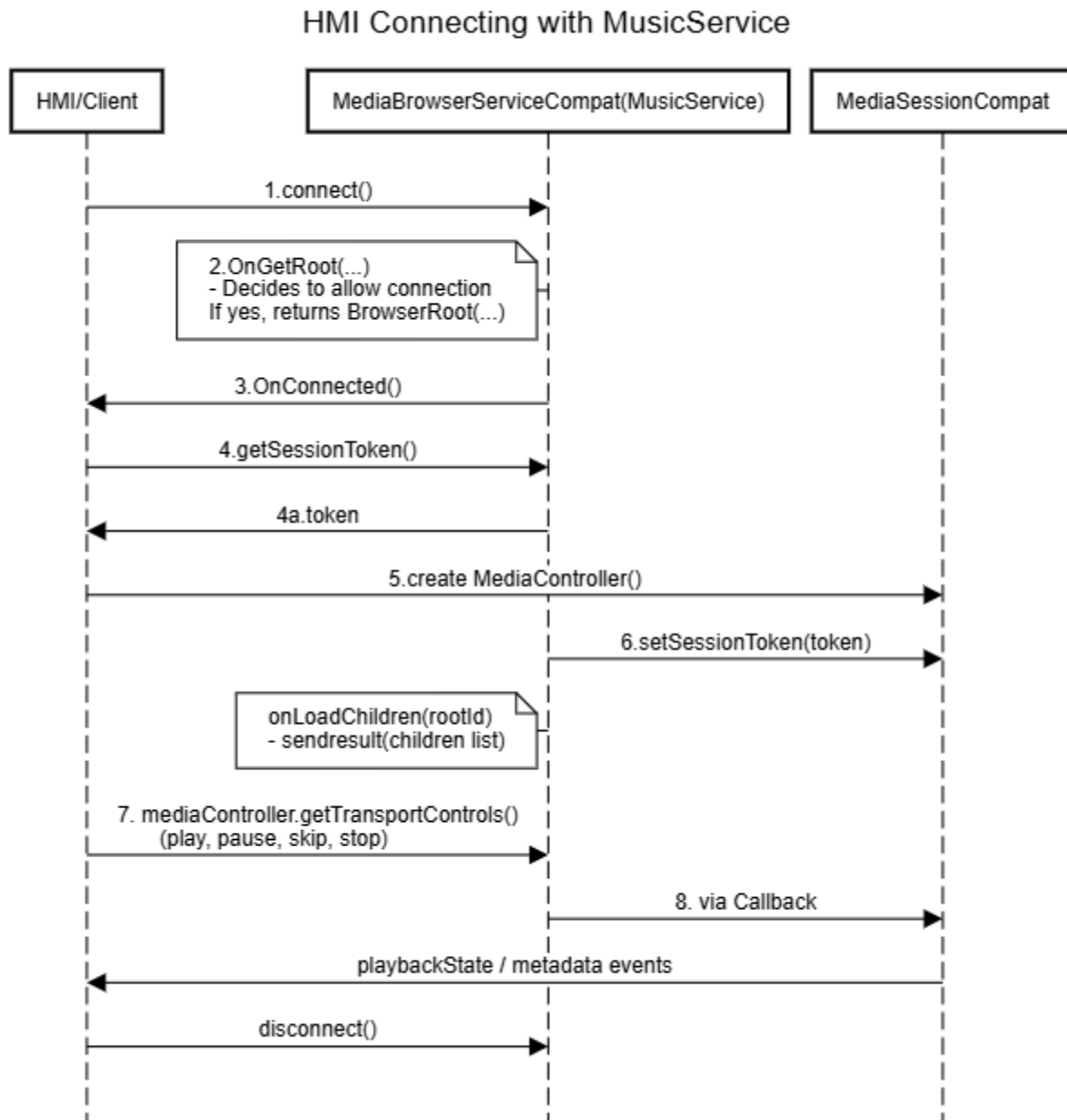


HMI to MusicService Connection

Here's a sequence diagram showing how an HMI (e.g. UI component or external client) connects with a MusicService that extends MediaBrowserServiceCompat.



Explanation of steps:

1. Client calls MediaBrowserCompat.connect()

The HMI / UI uses MediaBrowserCompat to initiate a connection to the MusicService (which extends MediaBrowserServiceCompat), supplying a ComponentName pointing to the service.

2. Service's onGetRoot(...) is invoked

The system calls the service's onGetRoot(clientPackageName, clientId, rootHints) to decide whether to allow the connection and what root content to expose.

3. Service returns a BrowserRoot

If allowed, onGetRoot() returns a BrowserRoot (with a root id and optionally extras). If it returns null, the client cannot connect.

4. Client's onConnected() callback fires

After successful root acquisition, MediaBrowserCompat.ConnectionCallback.onConnected() is invoked on the client side.

5. Client retrieves the MediaSessionCompat.Token

Using mediaBrowser.getSessionToken(), the client obtains the session token that the service had earlier set / exposed.

6. Client creates a MediaControllerCompat using the token

With that token, the client creates a MediaControllerCompat, which allows sending commands and receiving state/metadata.

7. Service sets the session token

In its onCreate(), the service must do setSessionToken(mediaSession.getSessionToken()) so that clients can retrieve it. Without this, clients can't control or observe the session.

8. Service's onLoadChildren(...) returns the list of media items

The service implements onLoadChildren(parentId, result) and either returns immediately via sendResult(...) or detach & later send. The children are MediaItems.

9. Client's onChildrenLoaded(...) is called

The client receives the list in its subscription's callback. It can then show UI, pick an item to play.

10. Client uses `MediaControllerCompat.getTransportControls()` to issue playback commands

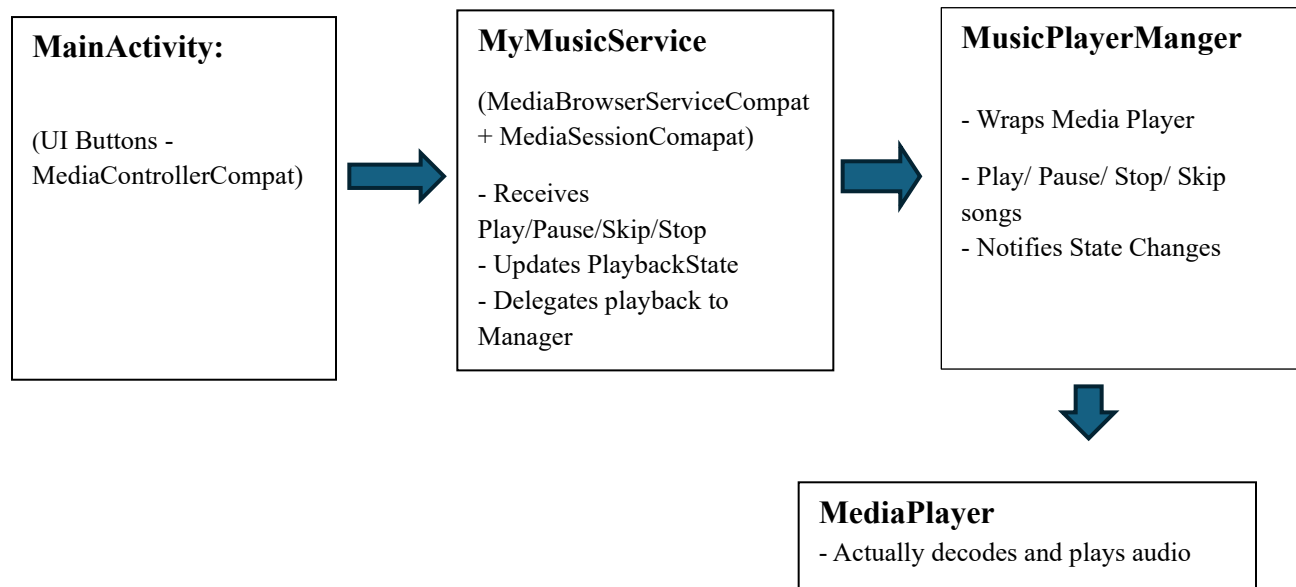
E.g. `playFromMediaId(...)`, `play()`, `pause()`, `stop()`, `skipToNext()`, `skipToPrevious()` etc. These are routed to `MediaSessionCompat.Callback` in the service.

11. Service pushes updates (metadata, playback state, queue, etc.)

The client registers a `MediaControllerCompat.Callback` to listen for metadata changes, playback state changes, etc., so the UI remains in sync.

12. Client eventually disconnects

E.g. when UI goes away; calls `mediaBrowser.disconnect()`. The service might stop itself if no clients remain, unless playback is ongoing.



GitHub Code Link: [Media/automotive/src at master · Aishwarya02082001/Media](https://github.com/Aishwarya02082001/Media/automotive/src)